Questions

Q1.

Figure 14 shows a footballer and a ball.

(Source: © GelgelNasution/Shutterstock)

Figure 14

The ball is stationary before it is kicked.

The mass of the ball is 0.43 kg.

The kick exerts a force of 2700 N on the ball.

The velocity of the ball just after it has been kicked is 36 m/s.

(i) Calculate the momentum of the ball just after it has been kicked.
State the unit for momentum.

(3)

momentum = ...

unit .....

(ii) Calculate the amount of time that the footballer's kicking foot is in contact with the ball.

(2)

time = ... s

(iii) Explain how Newton's third law applies to the interaction between the kicking foot and the football.

(3)

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(Total for question = 8 marks)

Q2.

Some students are investigating forces.

They use a newton meter to keep a trolley stationary on a sloping runway, as shown in Figure 3.

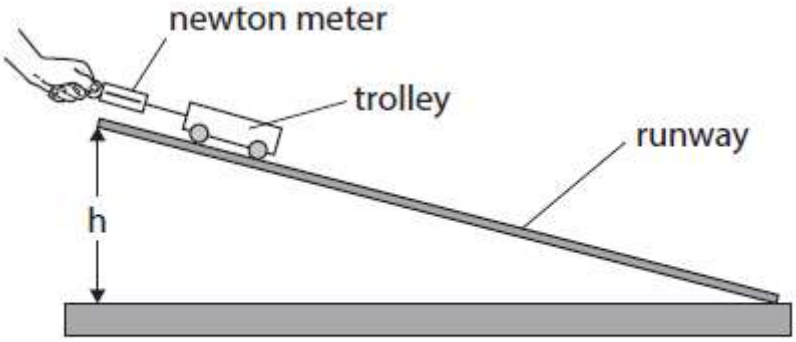

Figure 3

They measure the force needed to keep the trolley stationary on the sloping runway.

(i) They do this at four different places on the runway.

When the height, h, of the runway is 0.60 m, their four readings for force are

2.1 N 2.7 N 2.5 N 2.2 N

Calculate the mean value of these forces.

(1)

mean force = ... N

The students then measure values of the force, F, for several different values of h, the height of the runway.

Figure 4 is a graph of F against h.

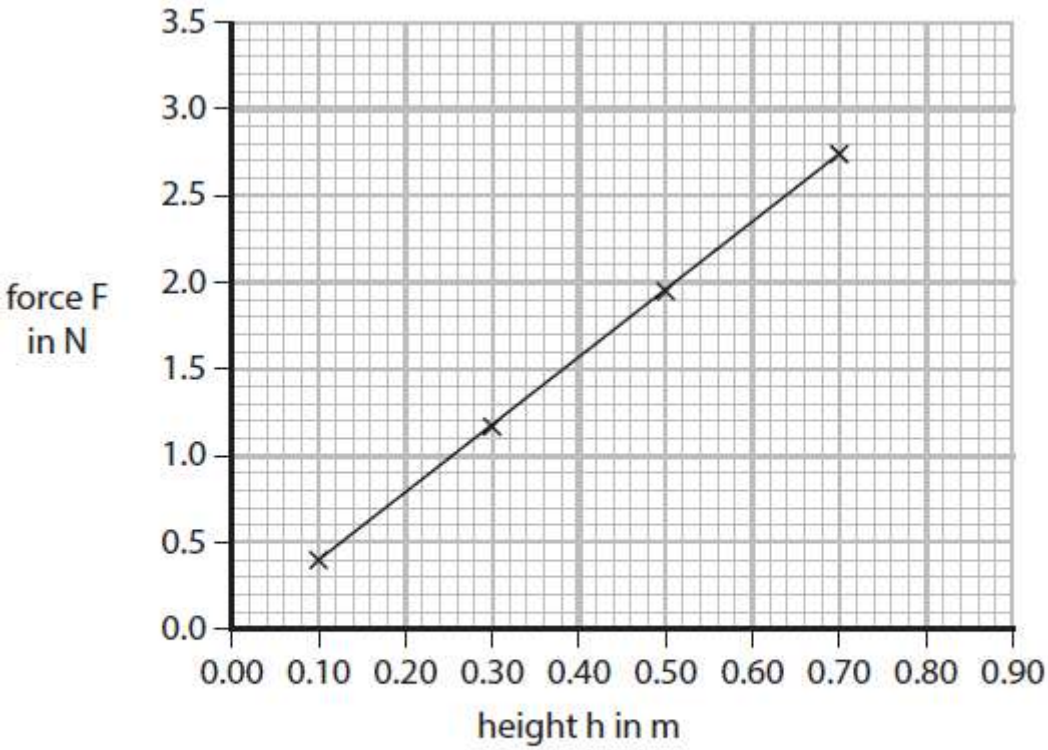

Figure 4

(ii) Use the graph in Figure 4 to determine the value of F when h = 0.80 m.

(2)

F = ... N

(iii) Determine the gradient of the graph in Figure 4.

(2)

gradient = ... N/m

(Total for question = 5 marks)

Q3.

Figure 2 shows the two forces acting on a beach ball as it falls through the air.

(Source: © Stepan Bormotov/Shutterstock)

Figure 2

The resultant of these two forces is 0.92 N downwards.

(i) The mass of the beach ball is 0.25 kg.

Calculate the acceleration of the beach ball as it falls.
Use the equation

$$F = m \times a$$

(3)

acceleration = ... m/s^2

(ii) Calculate the size of the air resistance force on the beach ball.

(1)

air resistance = ... N

(Total for question = 4 marks)

Q4.

A student has some cupcake cases.

One cupcake case is shown in Figure 2.

(Source: © Anton Starikov/Shutterstock)

Figure 2

The student drops a stack of cupcake cases with the base facing downwards, as shown in Figure 3.

(Source: © Elena Schweitzer/Shutterstock)

Figure 3

The speed of the falling stack of cupcake cases depends on the number of cupcake cases in the stack.

(i) The student also has a stop clock and a metre rule.

Describe an investigation to show how the speed of the falling stack of cupcake cases depends on the number of cupcake cases in the stack.

(4)

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(ii) A stack of cupcake cases has a mass of 0.005 kg.

Calculate the weight, in newtons, of the stack of cupcake cases.
Gravitational field strength = 10 N / kg

(2)

Use the equation

$$W = mg$$

weight = ... N

Figure 4 shows a cupcake case that is falling at a constant velocity.

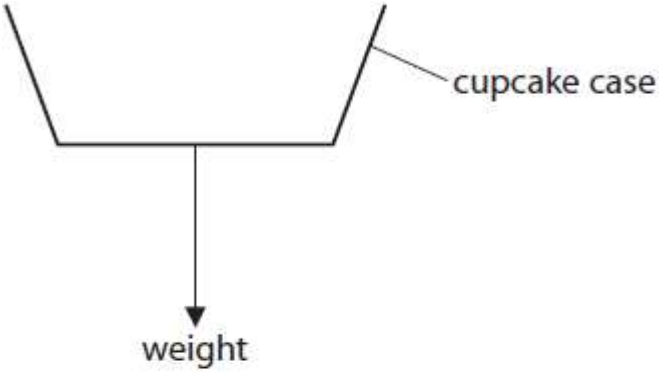

Figure 4

(iii) Draw an arrow on Figure 4 to show the force due to air resistance on the cupcake case.

(1)

(iv) State the value of the acceleration of the cupcake case when it is falling at a constant velocity.

(1)

.....

(Total for question = 8 marks)

Q5.

Figure 12 shows a different velocity/time graph.

This straight line graph can be represented by the equation

$$y = mx + c$$

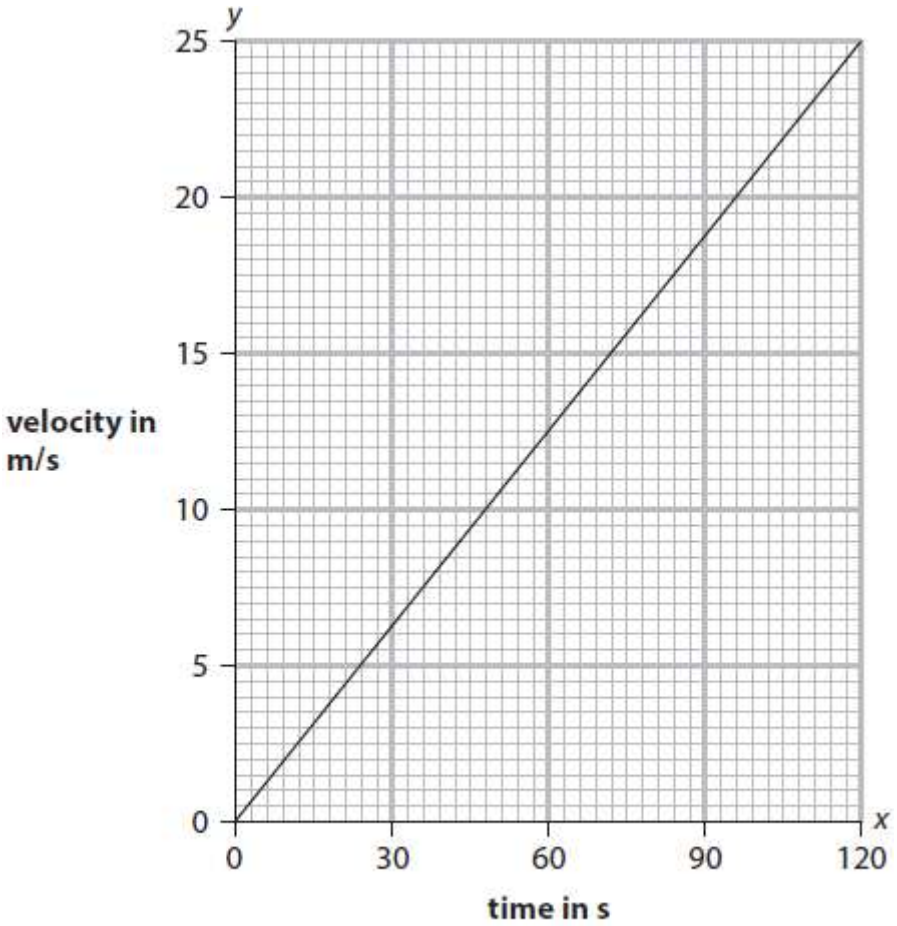

Figure 12

(i) Give the quantities that *x* and *y* represent in the equation.

(1)

x represents .....
y represents .....

(ii) Calculate the value of *m* from the graph in Figure 12.

(2)

m = ... m/s^2

(iii) State the value of *c* from the graph in Figure 12.

(1)

value of *c* = ...

(Total for question = 4 marks)

Q6.

Figure 10 shows a skier sliding down a slope.

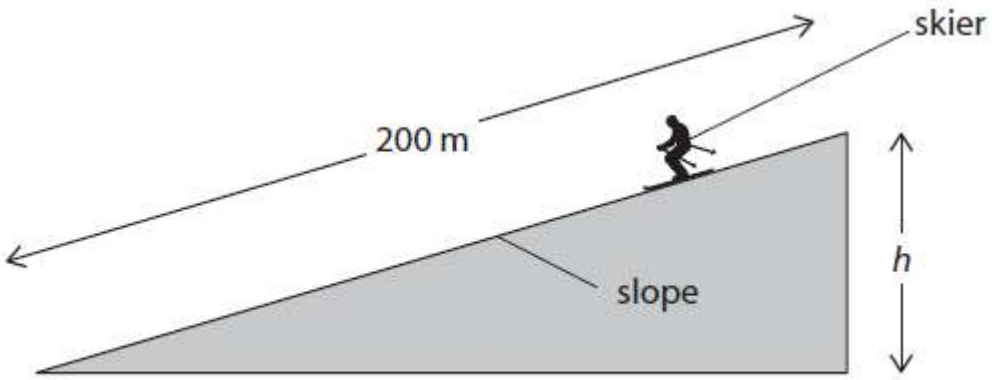

Figure 10

(i) The gravitational potential energy of the skier decreases by 39 000 J between the top and the bottom of the slope.

The mass of the skier is 70 kg.
Calculate the height, *h* of the slope.
The value of g is 10 N / kg.
Use the equation

$$\Delta GPE = m \times g \times h$$

(2)

h = ... m

(ii) The skier starts from rest at the top of the slope.

The velocity of the skier at the bottom of the slope is 27 m/s.
Using information from Figure 10, calculate the acceleration of the skier along the slope.

(3)

acceleration = ... m/s^2

(iii) When the skier slides from the top to the bottom of the slope, some energy is dissipated.

Explain how to determine the amount of energy dissipated.

(2)

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(Total for question = 7 marks)

Q7.

Figure 13 shows two ice skaters during a performance.

Figure 13

(i) The two ice skaters are travelling together in a straight line at 3.50 m/s.

Their total momentum is 371 kgm/s.
The man has a mass of 64.5 kg.
Calculate the mass of the woman.

(4)

mass = ... kg

(ii) Calculate the kinetic energy of the man.

(2)

kinetic energy = ... J

(Total for question = 6 marks)

Q8.

Figure 8 is a velocity/time graph showing a 34 s part of a train's journey.

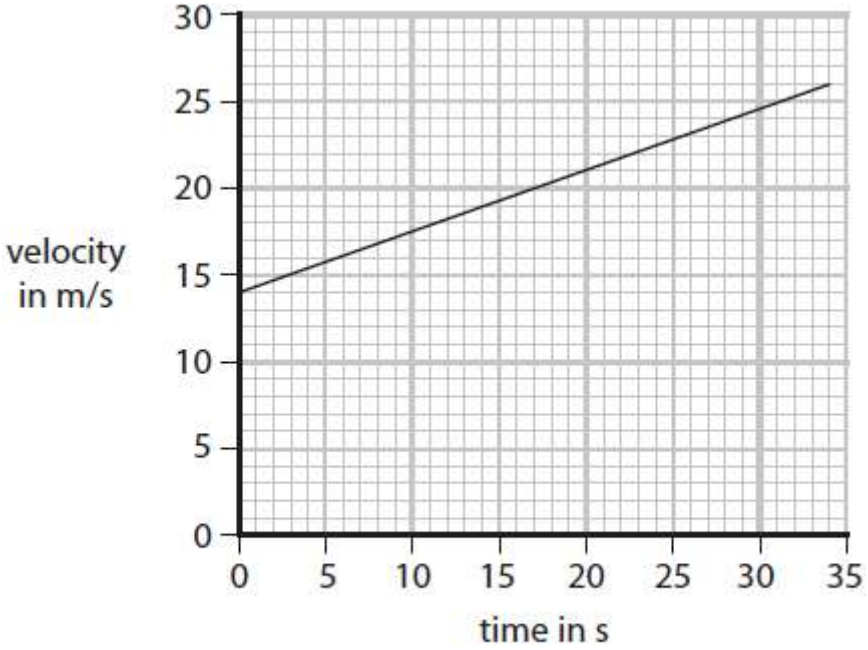

Figure 8

(i) Calculate the acceleration of the train in the 34 s.

Give your answer to an appropriate number of significant figures.

(3)

acceleration = ... m/s^2

(ii) Calculate the distance the train travels in the 34 s.

(3)

distance ..... m

(Total for question = 6 marks)

Q9.

Figure 8 is a velocity/time graph for a lift moving upwards in a tall building.

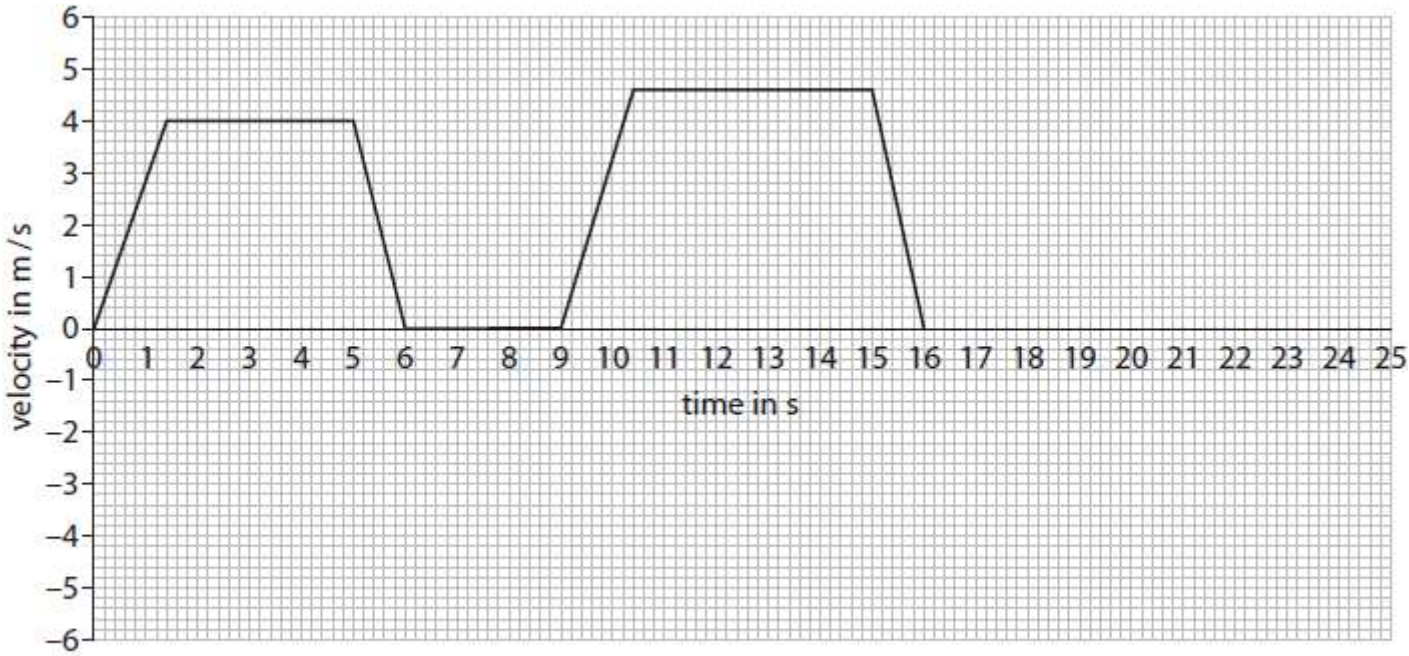

Figure 8

Use the graph in Figure 8 to determine the acceleration of the lift during the first 1.4 s.

(3)

acceleration = ... m/s^2

(Total for question = 3 marks)

Q10.

Figure 9 shows a racket and a tennis ball.

The tennis ball is travelling towards the racket at a velocity of 8.2 m/s.

The ball is hit back in the opposite direction at a velocity of 15 m/s.

The ball has a mass of 0.075 kg.

The ball is in contact with the racket for 12 ms.

(i) Calculate the average force exerted by the ball on the racket.
Use the equation

$$F = \frac{mv - mu}{t}$$

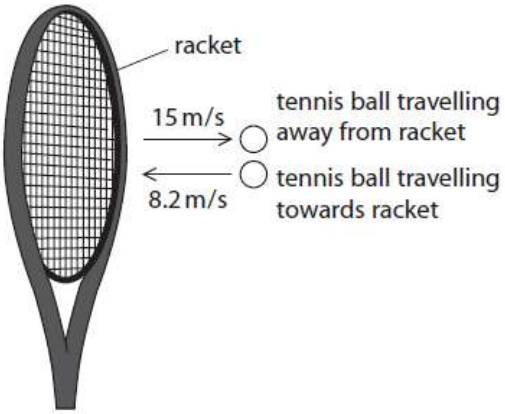

Figure 9

(3)

force = ... N

(ii) Describe how Newton's Third Law of Motion applies to the collision between the racket and the ball.

(2)

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(Total for question = 5 marks)

Mark Scheme

Q1.

Question number	Answer	Additional guidance	Mark
(i)	selection of (p =) m × v (1) evaluation (1) (momentum =) 15 unit (1) kg m/s	(momentum =) 0.43 × 36 allow numbers that round to 15 e.g. 15.48 allow 15.5 award 2 marks for the correct value without working kg m s^{-1} allow N s	**(3)** **AO2.1**

Question number	Answer	Additional guidance	Mark
(ii)	selection of force = $\frac{\text{change in momentum}}{\text{time}}$ (1) evaluation (1) 0.0056 (s) ALTERNATIVE METHOD selection of F=ma **and** a=(v-u)/t (1) evaluation (1) 0.0057 (s)	$2700 = \frac{15}{\text{time}}$ OR $2700 = \frac{\text{their (a)(i)}}{\text{time}}$ OR $(\text{time}) = \frac{15}{2700}$ OR $(\text{time}) = \frac{\text{their (a)(i)}}{2700}$ allow numbers that round to 0.0056 e.g. 0.005555 (using Δp=15) or numbers that round to 0.0057 e.g. 0.005733 (using Δp=15.48) allow ECF from their (a)(i) for 2 marks without working allow numbers that round to 0.0057 e.g. 0.00573 award full marks for the correct value without working	**(2)** **exp** **AO2.1**

Question number	Answer	Additional guidance	Mark
(iii)	explanation linking equal **and** opposite forces (1) (which are) force exerted by the foot (on the ball) (1) (and) force exerted by the ball (on the foot) (1)	 accept action and reaction are equal and opposite (which are) force exerted (by the foot) on the ball (1) (and) force exerted (by the ball) on the foot (1) award 2 marks for force on ball = force on foot award 3 marks for force on ball = force on foot but in opposite direction ignore references to other named forces and energy	**(3)** **AO2.1**

Q2.

Question Number	Answer	Additional guidance	Mark
(i)	2.4 (N) OR 2.375 (N) OR 2.38 (N)		**(1)** **AO2.2**

Question Number	Answer	Additional guidance	Mark
(ii)	straight line extended to meet h = 0.80 m axis (1) 3.1 (N) (1)	judge by eye allow 3.05 to 3.20 inclusive award full marks for the correct answer without working	**(2)** **AO2.2**

Question Number	Answer	Additional guidance	Mark
(iii)	select two points on the line (1) evaluation (1) 3.90 ± 0.10 (N/m)	allow triangle drawn on graph award full marks for the correct answer without working	(2) AO2.2

Q3.

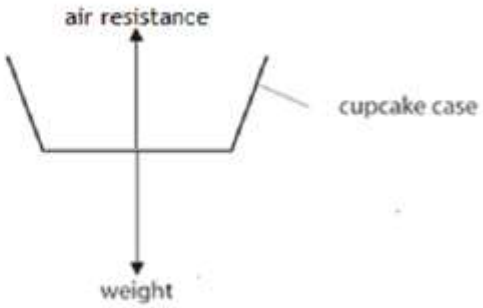
Question Number	Answer	Additional guidance	Mark
(i)	use of (F=)0.92 (N) (1) substitution and rearrangement (1) (acceleration) = $\frac{0.92}{0.25}$ evaluation (1) (acceleration) = 3.7 (m/s^2)	allow 0.92 x 0.25 OR $\frac{0.25}{0.92}$ allow MP2 for use of incorrect force giving $\frac{2.5}{0.25}$ OR $\frac{3.42}{0.25}$ OR $\frac{1.58}{0.25}$ OR $\frac{0.46}{0.25}$ allow values that round to 3.7 e.g. 3.68 allow values that round to 3.7 x 10^{-3} or 0.0037 for 2 marks (g instead of kg) allow for 2 marks an evaluation of 10(.0) OR 14 OR 13.7 OR 6.3(2) OR 1.8(4) (use of incorrect force) award full marks for the correct answer without working ignore negative values throughout if no other marks scored allow 1 mark for $a = \frac{F}{m}$	(3) AO2.1

Question Number	Answer	Additional guidance	Mark
(ii)	1.6 (N) OR 1.58 (N)		(1) AO3.1

Q4.

Question number	Answer	Additional guidance	Mark
(i)	A description to include any 4 from:		(4) AO1
	measure height (1)	allow 'keep same height' allow in this context hold against (fixed point on) metre rule	
	measure time of fall (1)	allow 'time it'	
	use (average) speed=distance ÷ time (1)		
	repeat with different number of cupcake cases in the stack/more cupcake cases (1)	accept cupcakes for cupcake cases	
	repeat **and** average time (of fall for each stack of cupcake cases) (1)		
	plot a graph (speed of fall against number of cupcake cases dropped) (1)		

Question Number	Answer	Additional guidance	Mark
(ii)	substitution (1) (W=)0.005 x 10 evaluation (1) 0.05 (N)	 5×10^{-2} (N) do not allow power of ten error award full marks for the correct answer with no working give full credit for use of g=9.8 or 9.81 N/kg	(2) AO2

Question number	Answer	Additional guidance	Mark
(iii)	 air resistance arrow (1)	Judge by eye any vertical upward arrow outside or inside the cupcake case ignore length of arrow arrow need not touch cupcake holder ignore label on arrow	(1) AO2

Question number	Answer	Additional guidance	Mark
(iv)	zero / there is none / 0 / it has no acceleration	ignore 'constant' ignore units	(1) AO2

Q5.

Question	Answer	Additional guidance	Mark
(i)	x represents **time** OR **t** y represents **velocity** OR **v**	both correct in correct place for 1 mark ignore units	**1** **AO1.1**

Question	Answer	Additional guidance	Mark
(ii)	attempt to find a gradient of the line that would give an answer between 0.18 and 0.24 (1) evaluation(1) 0.21 (m/s^2)	e.g. $\frac{12.5}{60}$ or $\frac{10}{42}$ values that are between 0.20 and 0.22 (m/s^2) e.g. 0.2083 do not allow fractions in the answer line for evaluation mark award full marks for the correct answer without working	**2** **AO2.1**

Question	Answer	Additional guidance	Mark
(iii)	0	zero	**1** **AO2.1**

Q6.

Question number	Answer	Additional guidance	Mark
(i)	rearrangement (1) $h = \frac{\Delta GPE}{m \times g}$ evaluation (1) (h) = 56 (m)	 $h = \frac{39\,000}{70 \times 10}$ allow numbers that round to 56 (m) e.g. 55.71 allow numbers that round to 57 (m) e.g. 56.85 (use of g=9.8 N/kg) award full marks for correct answer without working	**(2)** **AO3.2**

Question number	Answer	Additional guidance	Mark
(ii)	selection of $v^2 - u^2 = 2 \times a \times x$ (1) rearrangement (1) $a = \frac{v^2 (-u^2)}{2 \times x}$ evaluation (1) 1.8 (m/s^2) ALTERNATIVE METHOD calculation of time = 14.814 using time = distance / average speed (1) substitution (1) a (=(v-u) /t) = $\frac{27(-0)}{14.814}$ evaluation (1) 1.8 (m/s^2)	$27^2 (- 0^2) = 2 \times a \times 200$ allow $27^2 (- 0^2) = 2 \times a \times$ their (a)(i) $a = \frac{27^2(- 0^2)}{2 \times 200}$ allow $a = \frac{27^2(- 0^2)}{2 \times \text{their (a)(i)}}$ no ecf for evaluation allow numbers that round **down** to 1.8 (m/s^2) e.g. 1.8225 award full marks for correct answer without working.	**(3)** **AO3.2**

Question number	Answer	Additional guidance	Mark
(iii)	An explanation linking (measure) GPE at top/start and KE at bottom/end (1) energy (at top) subtract energy (at bottom) (1)	allow subtraction of any energies allow difference for subtraction award 2 marks for GPE at top subtract KE at bottom allow for 2 marks 39 000 – 25 515 OR 13485	(2) AO3.3a

Q7.

Question number	Answer	Additional guidance	Mark
(i)	substitution (1) 371 = (64.5+ m) x 3.5 rearrangement (1) m+64.5 = 371 / 3.5 evaluation of total mass (1) m+64.5 = 106 (kg) evaluation of woman's mass (1) m = 106-64.5 = 41.5 (kg)	full marks will be awarded for correct numerical answer without working	(4)

Question number	Answer	Additional guidance	Mark
(ii)	substitution (1) KE = ½ x 64.5 x 3.5^2 evaluation (1) 395 (J)	 allow answers which round to 395 e.g. 395.0625 full marks will be awarded for correct numerical answer without working	(2)

Q8.

Question number	Answer	Additional guidance	Mark
(i) CS4	attempt to use correct data from graph or equation (1) substitution (1) (a =) $\frac{26 - 14}{34}$ evaluation to 2 sf (1) 0.35 (m/s^2)	quoting $a = \frac{(\Delta) v}{t}$ or a = gradient (of line) 0.3529... scores mp1 and mp2 $\frac{26}{34}$ scores mp1 independent mark award full marks for correct answer without working.	(3) AO2

Question number	Answer	Additional guidance	Mark
(ii) CS4	attempt to calculate area under the line (1) calculates EITHER area of triangle OR area of rectangle (1) 204 (m) or 476 (m) evaluation (1) 680 (m)	accept count squares use of $v^2-u^2 = 2ax$ $x = \frac{v^2-u^2}{2a}$ allow ecf from b(i) award full marks for correct answer without working award 1 mark for final answer 408 (m)	(3) AO2

Q9.

	Answer	Additional guidance	Mark
	correct data point(s) seen (1) (accel =) $\frac{\Delta v}{t}$ (1) evaluation (1) 2.9 (m/s^2)	allow MP1 and MP2 in either order any data point(s) on the line e.g. (1.4,4) allow 'gradient' allow e.g. $\frac{4}{1.4}$ for 2 marks allow values that round to 2.9 (m/s^2) (e.g. 2.857...) award full marks for the correct answer without working	(3) AO3

Q10.

	Answer	Additional guidance	Mark
(i)	conversion of time to s (1) (t =) 0.012 **OR** 12×10^{-3} **OR** 1.2×10^{-2} substitution (1) $(F=)\frac{(0.075 \times -15.0) - (0.075 \times 8.2)}{0.012}$ OR $(F=)\frac{(0.075 \times 15.0) - (0.075 \times -8.2)}{0.012}$ OR $(F=)\frac{0.075 \times (15.0 + 8.2)}{0.012}$ evaluation (1) (-)150 (N)	substitution and conversion in either order ignore signs on velocity accept time to any power of ten for substitution mark $(F=)\frac{(1.125) + (0.615)}{0.012}$ 145 (N) scores 3 marks 145 (N) to any other power of ten scores 2 marks maximum	**(3)**

		145 (N) scores 3 marks 145 (N) to any other power of ten scores 2 marks maximum 42.5 (N) scores 2 marks maximum 42.5 (N) to any other power of ten scores 1 mark maximum 93.75 (N) or 51.25(N) 1.933 scores 1 mark maximum award full marks for correct answer without working	

	Answer	Additional guidance	Mark
(ii)	Any two from: (forces are) equal / same size (1) (forces are) opposite (direction) (1) (forces) act on different bodies (1) same type of force (1)	no marks awarded for answers in terms of energy (forces are) one to the left, one to the right one (force) acts on racket, one acts on ball both are contact forces if no other marks awarded, allow action and reaction (acting) for 1 mark	**(2)** **AO1**